

Brhanu Fentaw Znabu

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Research Interests

My research develops deep learning foundation models, large language models, and generative models for large-scale biological sequence analysis across genomes and proteins, with a focus on domain-adaptive pre-training, generative design, autonomous LLM agents for scientific analysis, and rigorous leakage-aware evaluation. I build systems end to end, from data curation through HPC training to open-source release, and apply them to predicting evolutionary trajectories and protein function.

Education

Ph.D. Biomedical Engineering Aug 2024 – Expected May 2029
University of Nebraska–Lincoln Lincoln, NE

• Advisors: Nicole R. Sexton, PhD (Nebraska Center for Virology) | Qiuming Yao, PhD (School of Computing)

M.Sc. Biomedical Science and Engineering 2021 – 2024
Gwangju Institute of Science and Technology (GIST) South Korea

- GPA: 4.0/4.5 | Advisor: Euiheon Chung, PhD
- Thesis: Deep behavioral phenotyping of diabetic neuropathy mouse model using open-field test

B.Sc. Biomedical Engineering 2014 – 2018
Jimma University Ethiopia

- GPA: 3.64/4.0 | Graduated with Distinction (Ranked 4th of 180)

Research Experience

PhD Researcher Aug 2024 – Present
Sexton Lab, University of Nebraska–Lincoln Lincoln, NE

- Built ArboFM, a domain-adapted transformer model via continued masked language model pre-training of DNABERT-2 (117M params) on 120K+ sequence windows across 362 species; achieved AP = 0.978 under species-grouped cross-validation preventing data leakage.
- Designed a leakage-aware evaluation framework quantifying 16-pp performance inflation from phylogenetic data leakage in cross-validation; applied gradient-based attribution and 6-mer enrichment analysis for model interpretability.
- Built a cross-domain transfer learning matrix across four biological families (6,052 sequences, 124K+ windows); demonstrated that learned representations are domain-specific (Jaccard $\approx$ 0), informing transfer learning strategy design.
- Performed retrospective temporal validation demonstrating the model detects distribution shifts in held-out future data, including within-class lineage discrimination (CLES = 0.898).
- Developed ViraPredict, a multi-modal forecasting framework integrating fitness landscape modeling, ESM-2 (650M params) zero-shot mutation scoring (F1 = 0.936), and LoRA-fine-tuned DNABERT-2 with temporal embeddings on 29,493 curated sequences across 130 countries.

Research Assistant Mar 2021 – Jun 2024
Neurophotonics Lab, GIST South Korea

- Developed unsupervised autoregressive and Hidden Markov models for temporal sequence modeling and state transition detection in behavioral time-series data.
- Benchmarked deep learning-based 3D pose estimation (DeepLabCut) against traditional 2D analysis using logistic regression on multi-dimensional behavioral features.

Open-Source Projects

- **RNA-seq Agent** (github.com/brhanufen/rnaseq-agent): Autonomous LLM agent that plans, writes, and runs Python for bulk RNA-seq analysis, self-corrects on errors, and returns grounded answers, shipped with a reliability benchmark and technical report: on a 10-task benchmark over GSE52778 (3 repetitions each) it solved every task with each reported number traceable to tool output, recovered from all induced errors, and used a median of 3 steps per task. The benchmark also documents that "how many genes are differentially expressed" has no single ground truth, ranging from 552 to $\sim$ 10,000 across pipelines.
- **TrustBinder** (github.com/brhanufen/trustbinder): Dual-oracle audit for de novo binder design, testing whether the score used to select generated binders survives an independent structure oracle. Boltz-2 and AlphaFold2 are uncorrelated on PD-L1 ($r = 0.08$, 100 designs) but strongly correlated on EGFR ($r = 0.64$), showing oracle concordance is target-dependent; on EGFR, where 55 of 402 designs have measured binding, gating on concordance lowers binder enrichment rather than raising it.
- **ProtDiff-ESM** (github.com/brhanufen/protdiff-esm): Discrete-diffusion protein language model conditioned on frozen

ESM-2 with classifier-free guidance toward a fitness target; training and reverse-diffusion sampler built from scratch, evaluated on ProteinGym with a leakage-aware harness whose ESM-2 baseline reproduces the public leaderboard (Pearson $r = 0.973$).

- **EnhancerDiff** (github.com/brhanufen/enhancerdiff): Discrete-diffusion model that designs cell-type-specific 200 bp enhancers (K562, HepG2, SK-N-SH) from the Gosai 2024 MPRA, validated against the Malinois and Enformer oracles.
- **OracleGap** (github.com/brhanufen/oraclegap): Oracle-independence and novelty audit for generative DNA design with Proto (Arc Institute); 93% of gradient-optimized designs transfer to an independent oracle (agreement $\rho = 0.865$) with no oracle-hacking detected.
- **SpliceConsensus** (github.com/brhanufen/spliceconsensus): Reproducible benchmark of splicing variant-effect predictors against the MFASS experimental assay (27,733 variants); Pangolin ranks best (AUROC 0.888) across five predictors, while 19% of splice-disrupting variants are missed by every tool, concentrated in the exon interior.
- **AbStab** (github.com/brhanufen/abstab): Leakage-aware benchmark for antibody thermostability prediction; showed random splits overstate generalization to novel antibodies and that a general PLM (ESM-2) relies on clonal leakage more than an antibody-specific model (AntiBERTy).
- **PerturbVAE** (github.com/brhanufen/perturbvae): Conditional VAE for single-cell perturbation-response prediction, evaluated under random, function-grouped, and combinatorial held-out splits on Norman 2019 and Replogle 2022 Perturb-seq data.
- **NativeReady** (github.com/brhanufen/nativeready): Sequence-based triage model and open benchmark for native mass spectrometry suitability (cluster-aware ROC-AUC 0.835 ± 0.029 , $n = 635$); shipped as a web tool, PyPI SDK, and CLI.

Publications

1. Ashiquzzaman A*, Lee E*, **Znabu BF***, Sakib AN, Chung G, Kim SS, Kim YR, Kwon H-S, Chung E. MoSeq based 3D behavioral profiling uncovers neuropathic behavior changes in diabetic mouse model. *Scientific Reports* 15, 15114 (2025). DOI: [10.1038/s41598-025-98184-9](https://doi.org/10.1038/s41598-025-98184-9)
2. **Znabu BF**, Atif Z. Interpretable deep learning-based multi-omics integration for prognosis in hepatocellular carcinoma. *bioRxiv*. 2026. DOI: [10.64898/2026.04.01.715980v1](https://doi.org/10.64898/2026.04.01.715980v1)
3. **Znabu BF**, Atif Z. NativeReady: an open benchmark and sequence-based triage model for native mass spectrometry suitability. *bioRxiv*. 2026. DOI: [10.64898/2026.05.03.722506v1](https://doi.org/10.64898/2026.05.03.722506v1)
4. **Znabu BF**, Atif Z, Devkota P, Alemu RH, K.C. J. A reproducible MFASS benchmark of splice-disruption predictors reveals a shared exon-interior blind spot. *bioRxiv*. 2026. DOI: [10.64898/2026.07.21.739871](https://doi.org/10.64898/2026.07.21.739871)
5. **Znabu BF**, Sexton NR. A genomic foundation model for predicting arbovirus epidemic emergence across RNA virus families. *Bioinformatics* (In preparation)
6. **Znabu BF**, Yao Q, Sexton NR. Deciphering the host-range grammar of orthoflaviviruses using foundation model embeddings: a leakage-aware evaluation framework. *PLOS Computational Biology* (In preparation)

Conference Presentations

- **ISMB 2026** (MLCSB COSI), poster accepted: Investigating Host-Range Determinants of Flaviviruses Using Foundation Model Embeddings: A Leakage-Aware Evaluation Framework. Washington, DC (2026). Plus symposium posters at UNL (2026) and the Nebraska Center for Virology (2025).

Technical Skills

Programming: Python, R, C++, MATLAB | **ML/DL:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, DNABERT-2, ESM-2, AntiBERTy | **Infrastructure:** GPU (H100/H200), HPC (Slurm), Git/GitHub

Methods: Foundation model pre-training (MLM), discrete diffusion, classifier-free guidance, variational autoencoders, LoRA fine-tuning, transfer learning, leakage-aware evaluation, gradient attribution, zero-shot prediction, PCA, UMAP | **Data:** NCBI GenBank, GISAID, ProteinGym, TCGA, GEO, Biopython, ESMFold, IEDB

Teaching Experience

Lecturer

Hawassa University

Aug 2019 – Jan 2021

Ethiopia

- Taught Biomechanics, Biomaterials, and Biophysics to undergraduate biomedical engineering students.

Awards & Grants

- Biswas Family Foundation Fast Grant (\$25,000, 2026) for *Forecasting Viral Emergence from Sequence: Generative Foundation Models for Arbovirus Surveillance*
- Nominated by UNL as one of four students for the Google PhD Fellowship 2026 in AI for Health
- Korean Government Scholarship for Master's Study, GIST (2020)
- Research Grant for Colostomy Device Development, Hawassa University (2020)
- Best B.Sc. Thesis Award, Ethiopian Science, Technology, and Innovation (2018)